

Title

Generalists link peaks in the shifting adaptive landscape of Australia's dragon lizards

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Abstract

The adaptive landscape is a flexible concept in biology which helps map entities onto an eco-evolutionary topography. A common practice is clustering species to identify adaptive peaks but the pathways that allow movement into new adaptive zones remain poorly understood. Here we integrate new sequence-capture (>5,000 nuclear markers) and multivariate morphological data to investigate the adaptive landscape of Amphibolurinae dragon lizards (Agamidae). Our well-supported, time-calibrated phylogeny includes over 360 specimens from 176 species, including iconic taxa such as the frilled-neck lizard, thorny devil, and bearded dragon. We show that the colonization of Australia was followed by rapid morphological expansion and facilitated by generalist forms. Ancestral dragons were likely arboreal, but generalist morphologies provided pathways to subsequent specialization. These findings reveal an evolutionary pathway for adaptive diversification in a major continental radiation.

Introduction

15 The adaptive landscape has become an indispensable tool for considering and
visualizing evolutionary spaces. In adapting Sewall Wright's (1932) genotypic fitness
landscape to a phenotypic space, GG Simpson (1944) provided evolutionary biologists
a shared vocabulary and conceptual bridge between micro- and macroevolution
(Arnold et al. 2001). At local scales (species or populations) landscapes help us to map
20 form to fitness. However, at broader scales (species, genera, families) landscapes (often
called morphospaces or functional trait spaces) tell us about the density and position
of peaks, and how communities or clades map to phenotypic space. This can be
valuable in determining the cost of moving between phenotypes, or comparing the
morphological limits of different clades. Importantly, phenotypic landscapes are
25 dynamic and can vary through time in their characteristics (Marshall 2014), dictating
the persistence and evolution of the species which they represent. As a result, we can
use data from those species to infer parameters of, and changes to, the landscape.

Despite how it is often visualized (2D, 3D), the adaptive landscape was never limited
30 to only two or three dimensions. Instead, landscapes can be multidimensional and here
the concept intersects with Hutchinson's (1957) idea of the niche hypervolume. In
multidimensional space we can assess the volume and density of individual peaks, probe
the depths of multidimensional valleys, or detect holes in the landscape hypervolume
(Blonder 2016). Higher dimensionality may also provide paths between peaks that
35 might not be identifiable with fewer dimensions. The complex topography of the
landscape, including holes, dictates viable paths from existing morphologies into new
ones. As such, the ideas of adaptive radiation and ecological opportunity have long
been entwined with Simpson's view of movement across the landscape (Yoder et al.
2010). With advances in computational power, development of more complex
40 comparative methods, and larger phenotypic datasets, we can begin to unravel
historical changes to the adaptive landscape and evolutionary responses that helped
develop new forms (Harmon et al. 2005). Importantly, we can ask which axes
facilitate/promote movement into new phenotypic space, and if changes to the
landscape are associated with shifts in traits such as ecology, habitat, geographic range,
45 and opportunity.

While evolutionary biologists have been visualizing the adaptive landscape and determining the relative heights of peaks since Simpson, comparatively little empirical attention has focused on the movement of lineages to populate new peaks and the evolution of the landscape through time (but see Marshall 2014; Burin et al. 2023). This is due in part to the necessity of a high dimensional dataset, well-resolved phylogeny, and an appropriate organismal group. Here we investigate patterns of adaptive landscape change and the role of ecology and expanding ranges in the morphological evolution of Australo-Papuan agamid lizards. Amphibolurine agamids consist of ~130 species with varied and diverse ecologies, habitat preferences, and highly imbalanced biogeographic richness. The majority of species (~110 spp.) are endemic to Australia, where their distributions span the continent's driest deserts and wettest rainforests. Their morphologies reflect these diverse habitats, ranging from forest dragons cloaked in spines, squat desert pebble mimics, and include iconic species such as the frilled-neck lizard (*Chlamydosaurus*) and thorny devil (*Moloch*). To match, they show a range of ecologies from cryptic arboreal species to sand dune burrowers. Unique ecomorphologies such as these provide an ideal opportunity to study the tempo and mode of evolution and the processes driving phenotypic change.

To address the evolution of the amphibolurine adaptive phenotypic landscape, we started by generating a sequence-capture dataset to reconstruct the phylogenetic relationships of Australo-Papuan dragon lizards. To investigate morphological diversity we collected an extensive phenotypic dataset of 19 linear measurements that summarize broad axes of variation across the head, body, limbs, and tail. We use phylogenetic comparative methods to assess the tempo and mode of trait evolution and from this present a picture of the evolution of amphibolurine dragon lizards. Across the dragon body, many traits show temporally and phylogenetically heterogeneous evolutionary histories. Those traits with evidence of varied evolutionary processes often exhibit large shifts in trait changes coincident with new morphologies. These morphological jumps are better described by biased directional changes which suggest selection, rather than increased evolutionary rates. Lastly, we identify that the diversity of forms of Australian dragons likely evolved from generalist ancestors which

fill space in the adaptive surface and act as intermediates connecting an otherwise patchy landscape.

80 Materials and Methods

Taxon Sampling

We assembled a sequence-capture dataset across 309 amphibolurine dragon lizards representing 120 of 136 currently recognized species, with a focus on Australian taxa (115 of 117 spp.). This sampling covers all 16 genera and many recognized subspecies
85 (Table S1). We included outgroup representatives from all agamid subfamilies and used tuatara (*Sphenodon punctatus*) and chicken (*Gallus gallus*) to root the tree (368 total samples).

Molecular Data Collection and Processing

90 Molecular sampling implemented the Squamate Conserved Loci (SqCL) kit (Singhal et al. 2017), which comprises ~5,000 ultra-conserved elements (UCEs), ~400 anchored hybrid enrichment loci (AHEs), and ~40 legacy genes commonly used in squamate molecular phylogenetics. Sample collection and laboratory preparation, including extraction, shearing, size selection, library generation, and hybridization capture were
95 completed as part of the Australian Amphibian and Reptile Genomics initiative (AusARG) under the Phylogenomics Working Group following the protocol outlined in Tiatragul et al. (2023). Raw sequence reads are available from the Bioplatforms Australia Data Portal (<https://data.bioplatforms.com/organization/ausarg>). We processed raw sequence data using the *pipesnake* workflow (Brennan et al. 2024). A
100 list of all software and versions used is included in the Supplement. *pipesnake* is a highly reproducible and flexible workflow that provides consistent results.

Phylogenetic Analyses

We estimated individual genealogies for our sequence-capture data (n=5441) under
105 maximum-likelihood in IQ-TREE2 allowing the program to assign the best fitting substitution model using ModelFinder (Kalyaanamoorthy et al. 2017), then perform 1,000 ultrafast bootstraps (Minh et al. 2013). We then estimated the species tree using the coalescent-consistent methods wASTRAL-hybrid and ASTRAL-IV, using IQ-

TREE2 gene trees as input. We used two ASTRAL methods to take advantage of the way wASTRAL-hybrid weights branch lengths and support values in estimating the species tree, and how ASTRAL-IV allows multiple individuals per species and estimates both internal and terminal branch lengths in substitutions-per-site (as opposed to coalescent units for internal branch lengths and static terminal branch lengths in wASTRAL). To quantify topological signal from individual gene trees we calculated gene concordance factors using IQ-TREE2 and focused on intergeneric relationships using a single representative for each genus.

Divergence Dating

To estimate divergence times among taxa we applied a series of fossil and secondary calibrations in MCMCTree (Rannala & Yang 2007) as outlined in the Supplement (Table S2). We started by trimming our ASTRAL tree down to a single representative for each ingroup species or subspecies for input. We reduced our molecular data to exonic markers in the AHE loci, then estimated raw genetic distances from alignments to use as a proxy for evolutionary rate. We removed the fastest 5% and slowest 5% of loci to avoid issues with extreme rate heterogeneity, removed third codon positions from all loci, then partitioned the remaining loci into three partitions using AMAS (Borowiec 2016).

For each partition we ran MCMCTree with *usedata* = 3 to get the approximate likelihoods and branch lengths using *baseml* (dos Reis & Yang 2011), then concatenated the out.BV files together. We then ran four replicate MCMCTree analyses on the gradient and Hessian (in.BV file; *usedata* = 2), each for 20k burnin generations before collecting 20k samples at a sampling frequency of 100 generations (2,020,000 total generations). We compared mcmc files for stationarity and convergence (ESS of all parameters > 200), combined them using logCombiner, and used this combined mcmc file to summarize divergence times on our tree (*print* = -1 in .ctl file). To validate our priors we ran an additional analysis (*usedata* = 0) to run explicitly from the prior calibrations and determine our effective priors for comparison against our posterior age estimates. We then plotted the applied priors against effective priors (priors as a result of multiple interacting priors from *usedata* = 0) and posterior estimates to ensure appropriate behavior of the MCMCTree analyses (Fig.S1).

Morphological Data Collection

We collected 19 linear measurements for 128 amphibolurine lizard species from 482
145 museum samples ($\bar{x} = 5$ per spp., min. = 1, max. = 8). Linear measurements were
collected by one author (IGB) using Mitutoyo digital calipers (product 500-763-20).
These measurements aimed to capture the gross morphology of the lizard body plan
and are distributed across the head, body, limbs, and tail (Fig.S2). Definitions of trait
measurements and adjustments are summarized in Table S3. Data processing and
150 analyses were carried out in R (R Core Team 2024).

The Evolution of Morphological Diversity

To identify the mode of individual morphological traits we fit a series of models ranging
from an unbiased random walk to rate-variable and trended. The most basic model—
155 Brownian Motion (BM)—allows phenotypes to evolve through incremental random
change. The most complex—*BayesTraits* V_4 's fabric model (Pagel et al. 2022)—also
follows a process of gradual change, but allows rates to shift at discrete points (changes
in *evolvability*, v) on the tree and allows traits to change rapidly along individual
branches through directed evolution (*trends*, β). Using *BayesTraits* V_4 we fit these
160 models and their derivatives (*fabric* with no v parameter, *fabric* with no β parameter)
and estimated their marginal likelihoods using a stepping-stone sampler (500 stones
each for 5,000 generations).

BayesTraits V_4 implements a reversible-jump MCMC sampler, which we specified to
165 run for 100 million generations with 10 million generations of burnin. We summarized
the output to check for convergence with a custom script
Scripts/processBTVarRates.R, and verified that all parameters had reached effective
sample sizes (ESS) > 200 . To verify model fit and consistency among runs we ran four
separate runs for each trait. Individual model outputs (...*VarRates.txt*) were
170 summarized using the *FabricPostProcessor* software and independent runs were
combined using *MergeFabric*. Significant evolvability (v) and trend (β) shifts had to
occur in all four runs.

To track trait values at internal nodes we logged values from the MCMC chain using the *AddTag* and *AddMRCA* features in *BayesTraits* by generating custom control files using functions defined in the Supplement. We also used this script to generate reasonable priors on the root state and rate by fitting a single rate BM model using *GEIGER* (Pennell et al. 2014). We summarized posterior estimates of ancestral trait values under varied models using custom scripts included in the Supplement. To track the evolution of dragon morphospace through time we reduced observed and estimated ancestral values to three principal component axes and fit multidimensional hypervolumes using *hypervolume* (Blonder et al. 2018). We then extracted the volume and density of hypervolumes at 0.5 million year intervals. To understand morphological distances between observed and estimated (ancestral) taxa/optima we measured Euclidean distances. We visualized functional morphological spaces using the package *funspace* (Pavanetto & Puglielli 2024)

To identify adaptive peaks across the multidimensional landscape (i.e., morphological regimes) we used an OU-based method that accounts for correlation among traits (*PhylogeneticEM*; Bastide et al. 2018). We used two different datasets as input. The first included all 19 non-overlapping morphological traits. Due to the strong correlation of some traits (Fig.S3) we generated a second dataset by reducing dimensionality through principal components analysis (PCA) and choosing the top six principal component axes which accounted for >90% of the cumulative variance, where each component explained >2%. We fit the models across a number of shift values ($K=0-15$) and a grid of 30 alpha values ($a=0-5$) to both datasets. For consistency, we fit a variety of additional schemes to test the reliability of our results, and those are explored further in the Supplement. After identifying the preferred number of optima (either 5 or 12; see Results; also called regimes) we used these as discrete characters for *RandomForest* (Liaw & Wiener, 2022) analysis to categorize ancestral taxa according to contemporary optima. Assigning species to ecological guilds is fraught with subjectivity so we relied on our morphological regimes to identify clusters of species that map loosely to existing ecological categorizations (e.g. arboreal, terrestrial) following the discussion of these species in Melville & Wilson (2019). RandomForest confusion (error) rates for almost all contemporary optima were sufficiently low

(<10%) and so we predicted the morphological regime of ancestral taxa (nodes) based on trait values estimated by the best fitting model for each trait.

Results

Molecular Data Collection and Processing

Sequence capture using the SqCL kit resulted in assembly of 5441 targets. New sampling recovered a median of 5118 loci, with a concatenated and aligned sequence length of >6 Mbp (Fig.S4). Additional sequence statistics are included in the Supplement.

Phylogenetic Analyses

Coalescent-consistent topologies estimated by wASTRAL-hybrid and ASTRAL-IV are well resolved with high support across most branches of the tree, including good resolution among genera and major clades (Fig.1; Fig.S5). Internal branches separating the Australian genera are short and consecutive rapid splits show high levels of incomplete lineage sorting resulting in low gene concordance factors (Fig.S6). This is notable in the positions of the *Moloch/Chelosania* and *Lophosaurus* clades and placement of *Tropicagama*. Additional areas of conflict are limited to interspecific placements of the *Ctenophorus fordi* group, the *Pogona minor* group, and the *Diporiphora pallida* group. In agreement with a recent study (Bruton et al. 2025) we find *Ctenophorus (Cryptagama) aurita*—a highly morphologically derived lineage—nested within *Ctenophorus*. Our phylogeny also supports the assignments of a number of frequently confused arid and savannah species, upholding the genera *Amphibolurus*, *Gowidon*, *Lophognathus*, and *Tropicagama*. We also provide strong evidence that unites *Rankinia*—the world’s highest latitude dragon—with *Pogona*. Beyond the Amphibolurinae, we also provide a strongly supported estimate of subfamilial relationships in the Agamidae. We recover a sister relationship between the Agaminae and Draconinae, which together are the sister to the Amphibolurinae. The next closest relatives are the Hydrosaurinae, followed by the Uromastycinae, and lastly the Leiolepidinae (Fig.1)

Divergence Dating

Amphibolurines appear sometime in the late Cretaceous (~75 mya), separating from their sister the Agaminae/Draconinae. The crown split among amphibolurines does not occur until much later, at the Eocene/Oligocene boundary ~34 mya. Cladogenetic events occur rapidly at the base of the amphibolurine tree, with the major splits among Australian groups all occurring between 30-25 mya. With the exception of the clade comprising *Chlamydosaurus*, *Lophognathus*, and *Amphibolurus*, all extant genera were established by the mid-Miocene ~15 mya. Rapid speciation events towards the base of the amphibolurine tree obscure some relationships, likely a result of quick successive splits in <600 ky (Fig.1).

Phenotypic Evolution

Of 19 morphological traits, 10 are best fit by the rate-variable and trended model from *fabric*, including the summary trait encapsulating size (Table S4). The remaining nine traits are best fit by a random walk model (Brownian Motion). Traits with greater variance and disparity are more likely to fit the *fabric* model (Fig.S7). Ancestral trait values for the 10 traits best fit by *fabric* were extracted and summarized from the mcmc output files. Ancestral trait values for the remaining nine traits were estimated under Brownian Motion. This provided a complete trait matrix for living and ancestral species which we used to quantify ancestral morphologies. Morphospace shows a heterogeneous pattern of expansion with a steep increase from 30-20 million years, before a period of little expansion from 20-10 million years, and another increase in volume from 10 million years to present (Fig.S8).

Multi-optima OU models identify either 5 or 12 morphological regimes. We refer to these regimes by the primary ecological habits of their members (arboreal, terrestrial, generalist) or the clade to which they belong (*Tympanocryptis*, *Moloch*), and consider how these morphological assignments map to ecology in the Discussion. Our multi-optima solutions are nested with the same 5 optima identified in the more conservative model and in the liberal 12 optima model. These optima correspond to arboreal taxa at the base of the tree (*Hypsilurus*, *Lophosaurus*, *Chelosania*), two separate terrestrial clades (*Tympanocryptis*, *Ctenophorus*), a generalist Australian semiarboreal group (*Amphibolurus*, *Gowidon*, *Intellagama*, *Tropicagama*, *Lophognathus*, *et al.*), and the unique *Moloch* (Fig.2; Fig.S9). More liberal models add optima which largely

270 correspond to genera (e.g. *Diporiphora*, *Pogona* + *Rankinia*) and outlier species
(*Chlamydosaurus kingii*, *Cryptagama aurita*, *Diporiphora superba*) and optimally
identify 12 peaks (Fig.S10). These clusters are largely visible when looking at raw
morphological distances among taxa (Fig.S11).

275 In both optima scenarios ($k = 5$, $k = 12$), RandomForest predictions suggest an
arboreal-type ancestor for the Amphibolurinae with some uncertainty ($k = 5$, 66%
assignment to arboreal; $k = 12$, 58%). The core Australian amphibolurine ancestor
(most recent common ancestor of *Intellagama* and relatives) is estimated to fit the
regime comprising semiarboreal generalists, however, its immediate ancestor is
280 estimated as fitting the arboreal regime. The amphibolurine MRCA is estimated to
have trait values most similar to *Chlamydosaurus*, *Intellagama*, and *Physignathus*
(Fig.S11,S12).

Discussion

The diversity of organismal forms in nature are not distributed evenly across the possible phenotypic space. Instead, observed phenotypes often cluster in discrete regions of morphospace, a result of genetic, functional, and/or developmental constraints (Deline et al. 2018). These clumps of morphologically similar organisms identify peaks in the adaptive landscape. Through time, the adaptive landscape may change as conditions vary and species go extinct, persist, and proliferate. The invasion of a new geographic region provides us the opportunity to observe how the adaptive landscape evolves, and to identify the tempo and mode with which new regions of the landscape become populated. Amphibolurinae dragon lizards are an ideal system to evaluate this because of their diversity of forms, age, and history. Our findings suggest that dispersal of dragon lizards into Australia facilitated a marked expansion in the morphological landscape. This exploration was driven first by a transition from an arboreal to a generalist ancestor and later into specialized ecomorphologies. Major morphological transitions were aided by focused evolution resulting in the rapid change of traits.

Phylogenomics of the Amphibolurinae

Dragon lizards (Agamidae) are a charismatic and diverse group of ~600 species found across Africa, Asia, and Oceania. They are diurnal and often brightly colored and territorial, making them visible and familiar. One subfamily, the Amphibolurinae, extends out of Asia into Australia and New Guinea. Here they reach their greatest ecomorphological extremes, ranging from burrowing species that live along the salt-crusted edges of desert lake-beds to semiaquatic species that patrol the eastern watercourses. While some species are familiar, such as the bearded dragons and frilled-neck lizards, to date, phylogenetic understanding of the group has been limited in genetic scope and focused within genera (Macey et al. 2000; Melville et al. 2001, Hugall et al. 2008; Shoo et al. 2008; Melville et al. 2018; Chaplin et al. 2020; Fenker et al. 2024). Our new sequence-capture dataset provides a well-supported estimate of the relationships among all 17 amphibolurine genera and more than 88% of described species.

315 Relationships among some (Streicher et al. 2016; Burbrink et al. 2020; Title et al. 2024)
or all (Welt & Raxworthy 2022) agamid subfamilies have been estimated from different
phylogenomic datasets, but have not previously been discussed. Our topology differs
from that of Welt & Raxworthy (2022) in the placement of the Leiolepidinae and
Uromastycinae at the base of the tree, and the stepwise (rather than sister) relationship
320 between the Hydrosaurinae and Amphibolurinae. The topology we present has
biogeographic implications, suggesting an Asian and not African origin for the group.
This is further supported by fossil evidence of the Priscagamidae which were stem-
acrodonts of east Asia, and the ~99 myo *Protodraco* from Myanmar (Wagner et al.
2021) which is potentially the oldest known acrodont. The debate over an Asian or
325 African origin of agamids is mirrored in other Australian squamate groups such as
pythons (Esquerre et al. 2020), monitor lizards (Brennan et al. 2021), and elapid snakes
(Keogh 1998). One possible explanation for differences in the subfamily topology is the
expanded molecular dataset presented here. Reanalyzing a UCE-only dataset however,
returns a topology identical to our SqCL estimate, suggesting other factors (e.g.
330 alignment parameters, substitution models) are likely causing the differences in
topology.

While the Amphibolurinae as a group date back to the Cretaceous, their living
diversity is less than half that age. The split between the sole Asian species
335 *Physignathus* and the remaining Australo-Papuan diversity occurred around the
Eocene/Oligocene transition, and was closely followed by rapid diversification over the
next ten million years (Fig.1). This is consistent with paleontological evidence from
Australia which suggests fossils assignable to *Intellagama* are relatively common in the
early-to-middle Miocene deposits of the Riversleigh World Heritage Area (Covacevich
340 et al. 1990). The rapid divergence of Australian species from their forest-dwelling
relatives in New Guinea resulted in a series of very rapid branching events which can
prove challenging for phylogenetic methods. These deep and rapid splits occur over
just a few hundred thousand years and highlight a period of dramatic ecomorphological
change. Following this period of radiation, the diversification of the group is far slower,
345 more homogeneous, and dictated by three major groups: *Ctenophorus* (41 spp.),
Diporiphora (28 spp.), and *Tympanocryptis* (26 spp.). This pattern of speciation slow-

down is consistent with theory around adaptive radiations and dispersal into new regions (Harmon et al. 2010; Schluter 2000; Simpson 1944; Yoder et al. 2010).

350 *Mapping Morphology to Ecology*

The adaptive landscape of amphibolurine dragons is dispersed and patchy, with peaks that largely correspond to recognized genera. Connecting these peaks are generalist semiarboreal forms that appear to facilitate the transition into more extreme morphological space and new ecologies. Obvious phenotypic peaks relate to strongly
355 arboreal (*Hypsilurus*, *Lophosaurus*) and terrestrial species (*Tympanocryptis*, *Ctenophorus*). However, intermediate forms that are adept at both running and climbing fill the morphological space and unite the more disparate peaks of the landscape (*Amphibolurus*, *Gowidon*, *Lophognathus*, *Tropicagama*, et al.). These generalist taxa are either lumped under a single broad adaptive peak (when $k=5$
360 optima; Fig.S9) or two peaks (when $k=12$ optima; Fig.S10). This is important because these intermediate morphologies appear common (~50% of species richness), are distributed across the entire continent in all habitat types, and provide a theoretical link between more extreme morphological forms. The idea of generalists as a bridge between morphotypes is not exclusively a reflection of how we generate morphospaces.
365 In dragon lizards this process is informed by the branching pattern of the phylogeny. Specialist forms often descend from morphologically intermediate ancestors, and in some cases are nested deeply within generalist clades. For example the stout rock-dwelling *Diporiphora bennetti* and gracile hummock grass living *Diporiphora adductus* likely both evolved from a more typical amphibolurine morphology seen in their closest
370 relatives.

Linking morphology to ecology however, is not always so clear. This may be the result of a many-to-one mapping where varied morphologies enable the same ecology (Wainwright et al. 2005; Moen 2019). For example, taxa such as *Lophosaurus*,
375 *Chelosania*, and *Diporiphora superba* are all obligately arboreal and highly specialized dragons, however, these taxa map to either two or three discrete adaptive zones (Fig.2) and use arboreal spaces differently. *Diporiphora superba* is an extremely gracile species with a long tail, small head, and long slender limbs that it uses to move through shrubs. *Lophosaurus* species similarly have long tails and limbs, but have large heads

and deep, laterally compressed bodies and generally perch vertically on tree trunks. *Chelosania* also show a narrowing of the body but have short sturdy limbs and a relatively short tail for a species that is almost never found outside of the canopy where they often use horizontal branches. In this situation, convergence on a specialized lifestyle is met with only incomplete morphological similarity. This may be due to a number of reasons including phylogenetic inertia resisting selection (Blomberg & Garland 2002). This idea is relevant when considering the ecology of the ancestral amphibolurine. Our analyses suggest this ancestor was likely morphologically similar to the arboreal *Lophosaurus* and *Hypsilurus*, but also to *Physignathus* and *Intellagama*—two species that are considered semiaquatic (Fig.SX12). These last two species are similar enough that they were considered congeneric for over 150 years and are often called “water dragons”. Despite their preferred habitat, they fit most morphological descriptors of a tree-living dragon. We flag this to note the imperfect mapping of morphology to ecology and uncertainty in estimating ancestral states.

The Evolving Adaptive Landscape

The transition from wet closed-canopy rainforests in Asia and New Guinea towards new and different environments in Australia presents a prime case for ecological opportunity (Collar et al. 2010; Gray et al. 2019a). Among the newly available habitats, dragons encountered sparse open forests, grasslands, sandy and stony deserts, and rocky escarpments, which are all but absent across Malesia. By expanding their geographical range, dragon lizards also extended the adaptive landscape to enable new forms that took advantage of these habitats (Yoder et al. 2010). This extension provided opportunity for new peaks located in distant regions of morphospace. To traverse that space however, required deterministic trends in trait evolution that suggest selection on individual or suites of traits (Figs.3,4). In dragon lizards, this may have first resulted in a transition from an arboreal form to a generalist form. Our analyses indicate that this occurred rapidly and dramatically, resulting in a multivariate shape change dozens of times greater than expected (Fig.3 highlight 5). The resulting form likely had shorter limbs and tail, a wider body, and shorter snout as it spent less time in the trees (Fig.4). From generalist forms amphibolurines were able to further explore morphological extremes. This includes repeated transitions

towards small-bodied terrestrial forms (*Ctenophorus aurita*, *Rankinia*,
Tympanocryptis), (Fig.4), and moves back towards elongate arboreal forms
415 (*Diporiphora superba*, *Chlamydosaurus*). In this way, generalists negate the need to
cross wide fitness valleys in the adaptive landscape. Instead, species can explore across
ridges or broad fitness peaks to reach distant morphospaces. This provides more
evidence that transitions across the landscape from specialist to specialist are
uncommon and likely require an intermediate form.

420 In our investigation across the dragon lizard body plan, we identify varied modes of
evolution among and within traits. In the simplest sense, the evolution of dragon
morphology is dictated by both a gradual diffusion process away from stable peaks and
infrequent shifts in the evolutionary rate or trait mean. Perhaps most exciting about
425 these findings is how rare these shift events are. To explain the disparate morphological
landscape of dragons less than 10% of branches need multivariate trait change that
exceeds Brownian expectations. We can see this when looking across the morphospace,
where discrete regions are heavily occupied (Fig.1). These high density areas indicate
slow evolutionary rates or constrained evolution (i.e. Ornstein Uhlenbeck), as has been
430 previously identified in *Diporiphora* (Smith et al. 2010). Stable peaks populated by
many closely-related species highlight diffusion as the most frequent pathway for
change among species. In comparison, rare evolutionary shifts are concentrated around
changes among adaptive zones. These shifts are nested and temporally staggered
resulting in a dynamic adaptive landscape that does not reflect the archetypical tempo
435 or mode of an adaptive radiation. This complex mosaic of evolutionary modes helps to
reconcile Simpson's idea of quantum evolutionary change (few, extreme changes) with
Darwin's ideas of gradualism (frequent, small changes) (Simpson 1944; Charlesworth
et al. 1982).

440 The morphological evolution of dragon lizards has been a heterogeneous process,
varying temporally, phylogenetically, and across traits. Broadly, dragons have gone
through periods of early and late expansion in the morphological hypervolume,
bookending a period of little innovation during much of the Miocene (Fig.S8). Looking
across all species, we identify that the primary axes for variation are through change
445 in absolute size and tail length (Fig.S7). This is immediately apparent when comparing

large tree dragons (*Hypsilurus magnus*: SVL >200 mm, tail length >600 mm) to pebble-mimicking dragons (*Ctenophorus aurita*: SVL <40 mm, tail length <35 mm; *Tympanocryptis cephalus* group: SVL <60 mm, tail length <90 mm). Elaboration in size and tail length beyond other traits is a pattern seen in other squamate clades (Brennan et al. 2024; Meiri 2010), but dramatic changes are not limited to these axes. Instead, amphibolurines show exceptional variation in many traits and greater than expected diversity relative to other iguanians (Gray et al. 2019a; Gray et al. 2019b). Most trait trends appear associated with habitat type (Thompson & Withers 2005; Collar et al. 2010), such as broader heads and wider bodies in terrestrial groups or longer limbs and snouts in arboreal species. But others are harder to quantify and likely associated with defense or communication, like the frill of *Chlamydosaurus* or fixed and erectable crests in many species. Harder yet to explain is the tremendous novelty of *Moloch*, a lizard commonly called the thorny devil for the keratinous spines that cover its body. Despite their familiarity, *Moloch* remain enigmatic in many ways. They likely pre-date the Australian deserts in which they wander (Pepper & Keogh, 2021). They exclusively eat ants. And they have undergone such a profound evolutionary modification that they are nearly unrecognizable as agamids. In expanding into this new adaptive zone, the thorny devil has stretched the morphological space, representing just one of many exciting peaks in the landscape of dragon lizards.

Conclusion

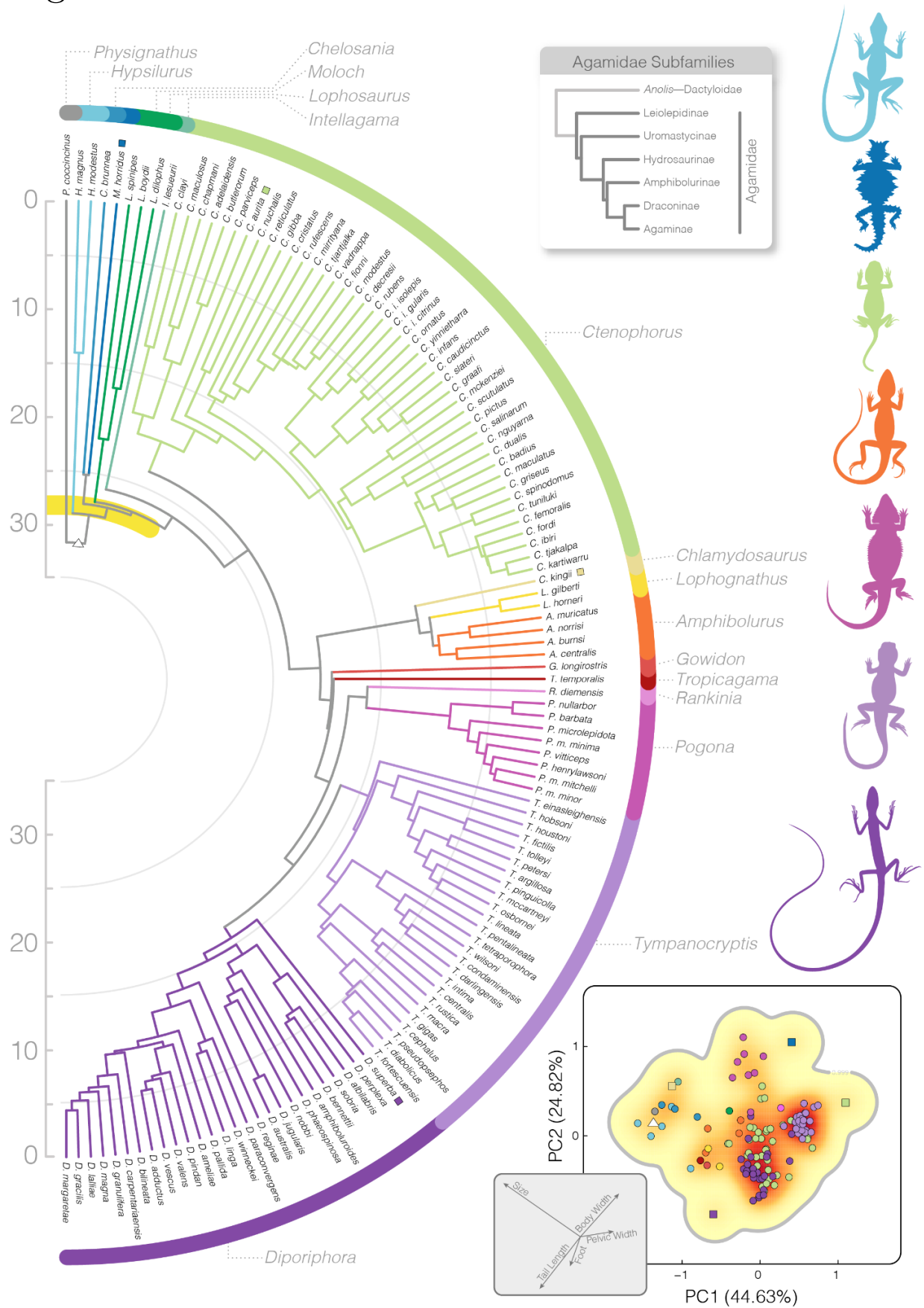
The adaptive landscape is a popular metaphor for linking form to function and contextualizing organismal diversity. For a long time we have focused on studying the peaks, but it has been difficult to imagine how species have traversed the landscape to establish new adaptive zones. Our investigation into the evolution of the morphological landscape in amphibolurine dragon lizards emphasizes the importance of generalist forms in connecting discrete peaks. While the ancestral amphibolurine was likely an arboreal dragon, the diversity of Australian forms were only realized following strong directional changes to a labile generalist form. As others have noted (Collar et al. 2010; Gray et al. 2019a; Gray et al. 2019b), this was likely driven by ecological opportunity in the form of a range of new habitats after landing in Australia. As a result, Australian

dragons represent one of the continent's most impressive adaptive radiations. Importantly, this system shows how the adaptive landscape itself can change and grow, expanding its borders and altering its topography in response to new opportunities.

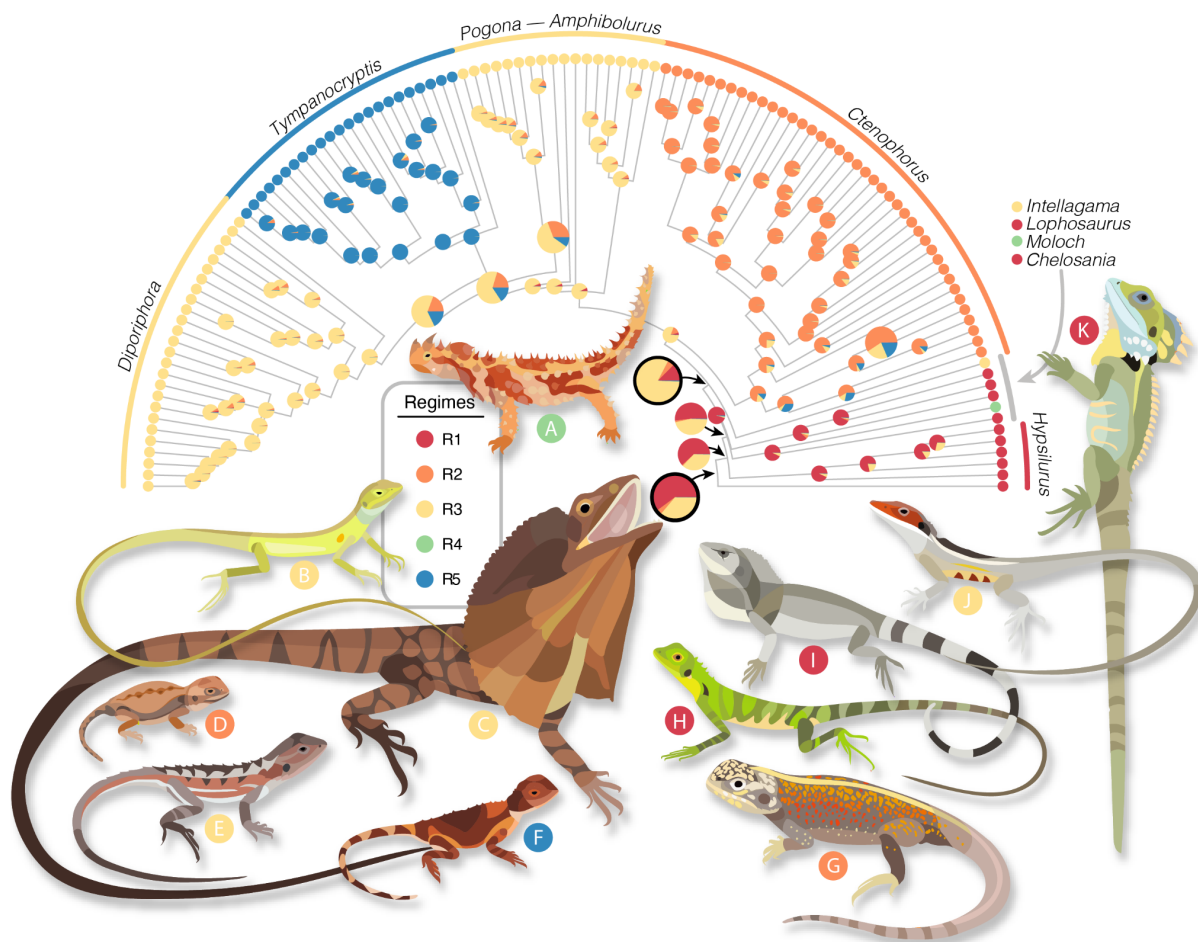
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Figures



495 Figure 1. Species-level timetree of the Amphibolurinae, color-coded by genus. The
ASTRAL species tree time-calibrated with MCMCTree shows a rapid Oligocene
divergence of most major amphibolurine groups. A yellow band highlights the series of
rapid splits early in the history of this group. Inset bottom right shows the spread of
species in two-dimensional morphospace (also colored by genus) and primary axes of
500 variation. *Diporiphora* (dark purple) and *Tympanocryptis* (light purple) highlight
dense clusters of morphospace, contrasting with morphological outliers indicated by
squares in the morphospace and on the tree (*D. superba*, *M. horridus*, *C. aurita*). The
estimated root morphotype is indicated by a triangle. Inset top right shows
relationships among agamid subfamilies and color-coded dragon drawings at right (not
505 to scale) highlight some of the morphological diversity of this group. White circles on
the tree denote local posterior probabilities <0.95 .



510 Figure 2. The adaptive morphological zones of amphibolurine dragons highlight a
 diversity of forms. Colored regimes were identified under a multi-optima OU model in
PhyloEM and applied to ancestral taxa using *randomForests*. The tree shows the
 preferred 5 regime model estimated from all 19 size corrected morphological traits and
 likelihoods for ancestors as estimated by *randomForest*. Two regimes (blue, orange)
 515 map to obligately terrestrial groups. The red regime highlights primarily arboreal
 species and the morphologically similar *Physignathus*. Yellow indicates a diverse clade
 of primarily generalist species. The amphibolurine most recent common ancestor
 (MRCA) is estimated as a *Hypsilurus*/*Lophosaurus*-like arboreal dragon and the
 Australian MRCA is estimated as a semiarboreal generalist lizard (both indicated with
 520 black outlined pie). Illustrated species are (A) *Moloch horridus*, (B) *Diporiphora*
superba, (C) *Chlamydosaurus kingii*, (D) *Ctenophorus aurita*, (E) *Rankinia diemensis*,
 (F) *Tympanocryptis pseudopsephos*, (G) *Ctenophorus reticulatus*, (H) *Hypsilurus*
modestus, (I) *Chelosania brunnea*, (J) *Gowidon longirostris*, (K) *Lophosaurus boydii*.

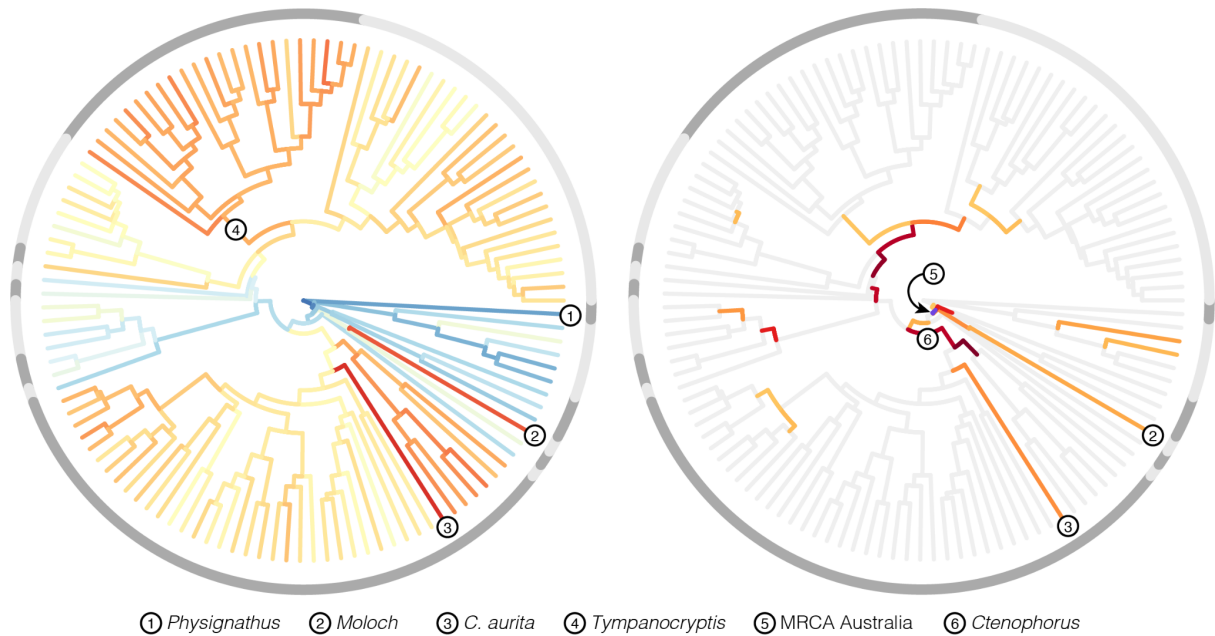


Figure 3. Major shifts in morphologies occur as directional evolutionary changes that are greater than expected. Left, morphological innovation across the dragon tree, with warm colors indicating greater multivariate Euclidean distance from the amphibolurine most recent common ancestor, and cooler colors indicating greater similarity. Grey shading in outer circles delineates genera. Right, the position and strength of directional changes that exceed multivariate shape change under the null evolutionary model. Circle number 5 indicates the branch leading to the Australian radiation of dragon lizards and denotes a 26-fold increase over the expected amount of morphological change. This very short branch accounts for a paradigm shift in the evolution of amphibolurine lizards away from an arboreal ancestor towards a more generalist morphology.

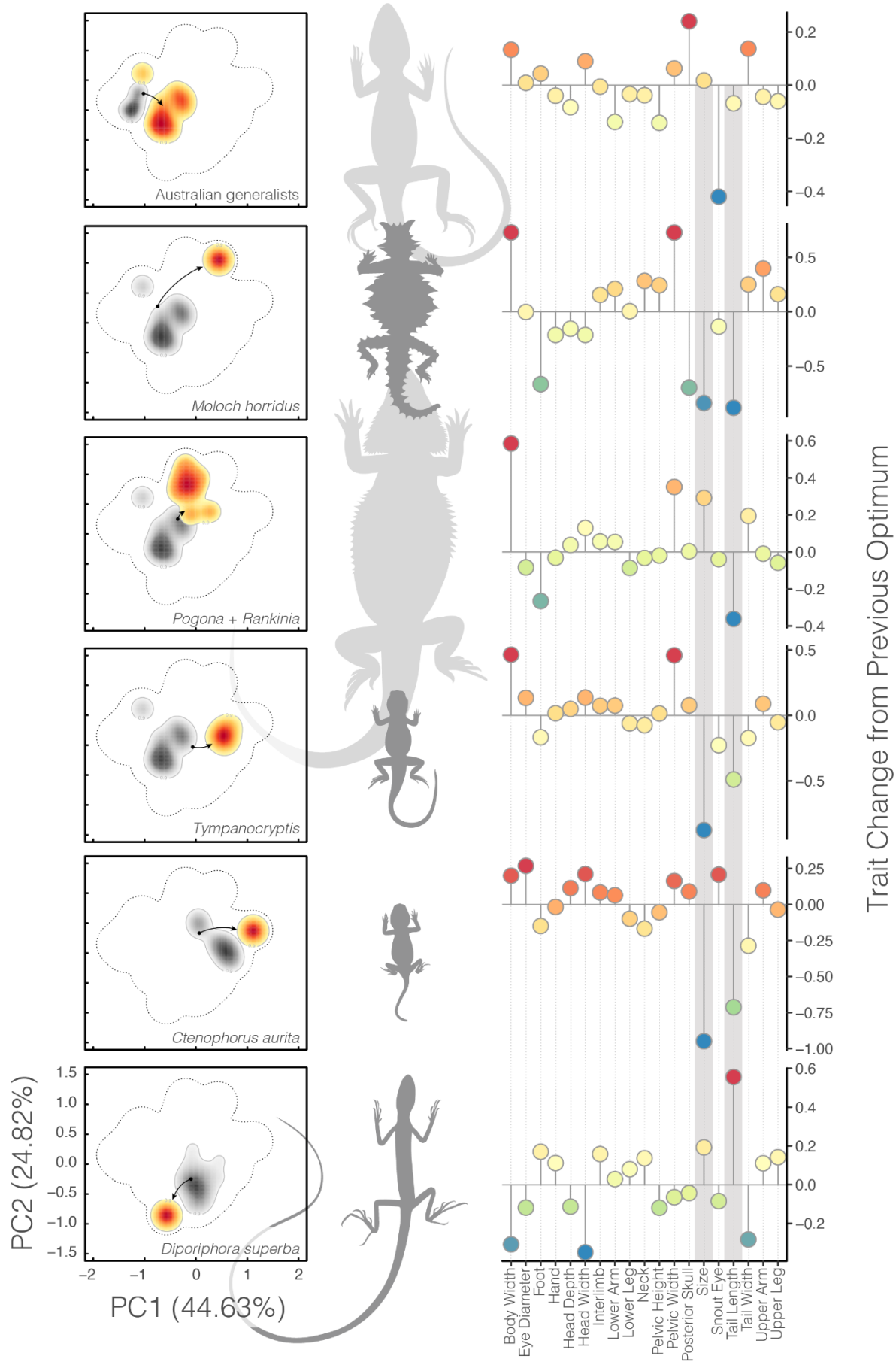


Figure 4. Movement among optima occurs via changes in different suites of traits. Plots in the left column visualize the shift from the immediately ancestral optimum (grey shading) to a new optimum (warm shading) represented in two dimensional space. Lizards at center illustrate the morphological extremes of new optima. Plots at right indicate how individual traits changed in evolving from the previous optima to current. Grey bars highlight the two most variable traits: size (a composite of the geometric mean of all trait measures) and tail length.

References

- 550 Arnold, S. J., Pfrender, M. E., & Jones, A. G. (2001). The adaptive landscape as a conceptual bridge between micro-and macroevolution. *Genetica*, 112, 9-32.
- Bastide, P., Ané, C., Robin, S., & Mariadassou, M. (2018). Inference of adaptive shifts for multivariate correlated traits. *Systematic Biology*, 67(4), 662-680.
- Blomberg, S. P., & Garland Jr, T. (2002). Tempo and mode in evolution:
555 phylogenetic inertia, adaptation and comparative methods. *Journal of Evolutionary Biology*, 15(6), 899-910.
- Blonder, B. (2016). Do hypervolumes have holes?. *The American Naturalist*, 187(4), E93-E105.
- Blonder, B., Morrow, C. B., Maitner, B., Harris, D. J., Lamanna, C., Violle, C.,
560 Enquist, B. J., & Kerkhoff, A. J. (2018). New approaches for delineating n-dimensional hypervolumes. *Methods in Ecology and Evolution*, 9(2), 305-319.
- Borowiec, M. L. (2016). AMAS: a fast tool for alignment manipulation and computing of summary statistics. *PeerJ*, 4, e1660.
- Brennan, I. G., Lemmon, A. R., Lemmon, E. M., Portik, D. M., Weijola, V., Welton,
565 L., ... & Keogh, J. S. (2021). Phylogenomics of monitor lizards and the role of competition in dictating body size disparity. *Systematic Biology*, 70(1), 120-132.
- Brennan, I. G., Singhal, S., & Al Bkhetan, Z. (2024). *pipesnake*: generalized software for the assembly and analysis of phylogenomic datasets from conserved genomic loci. *Bioinformatics*, 40(5), btae195.
- 570 Brennan, I. G., Chapple, D. G., Keogh, J. S., & Donnellan, S. (2024). Evolutionary bursts drive morphological novelty in the world's largest skinks. *Current Biology*, 34(17), 3905-3916.
- Bruton M., Melville J., Sumner J., Porter J., Mahony S. & Doughty P. (2025). The last piece of the puzzle: Detection of an extremely rare lizard in the Kimberley completes the phylogeny of Australian dragons (Agamidae). *Australian Journal of Taxonomy* 93: 1–15. doi: <https://doi.org/10.54102/ajt.zoa0>
- 575 Burbrink, F. T., Grazziotin, F. G., Pyron, R. A., Cundall, D., Donnellan, S., Irish, F., ... & Zaher, H. (2020). Interrogating genomic-scale data for Squamata (lizards, snakes, and amphisbaenians) shows no support for key traditional morphological relationships. *Systematic biology*, 69(3), 502-520.
- 580 Burin, G., Park, T., James, T. D., Slater, G. J., & Cooper, N. (2023). The dynamic adaptive landscape of cetacean body size. *Current Biology*, 33(9), 1787-1794.
- Chaplin, K., Sumner, J., Hipsley, C. A., & Melville, J. (2020). An integrative approach using phylogenomics and high-resolution X-ray computed tomography
585 for species delimitation in cryptic taxa. *Systematic Biology*, 69(2), 294-307.
- Charlesworth, B., Lande, R., & Slatkin, M. (1982). A neo-Darwinian commentary on macroevolution. *Evolution*, 36(3), 474-498.

- Collar, D. C., Schulte, J. A., O'meara, B. C., & Losos, J. B. (2010). Habitat use affects morphological diversification in dragon lizards. *Journal of Evolutionary Biology*, 23(5), 1033-1049.
- 590 Covacevich, J., Couper, P., Molnar, R. E., Witten, G., & Young, W. (1990). Miocene dragons from Riversleigh: new data on the history of the family Agamidae (Reptilia: Squamata) in Australia. *Memoirs of the Queensland Museum*, 29(2), 339-360.
- 595 Deline, B., Greenwood, J. M., Clark, J. W., Puttick, M. N., Peterson, K. J., & Donoghue, P. C. (2018). Evolution of metazoan morphological disparity. *Proceedings of the National Academy of Sciences*, 115(38), E8909-E8918.
- dos Reis, M. and Yang, Z. (2011). Approximate likelihood calculation for Bayesian estimation of divergence times. *Molecular Biology and Evolution*, 28:2161-2172.
- 600 Esquerré, D., Donnellan, S., Brennan, I. G., Lemmon, A. R., Moriarty Lemmon, E., Zaher, H., ... & Keogh, J. S. (2020). Phylogenomics, biogeography, and morphometrics reveal rapid phenotypic evolution in pythons after crossing Wallace's line. *Systematic Biology*, 69(6), 1039-1051.
- Fenker, J., Melville, J., & Moritz, C. (2024). Dragons in the tropics—Phylogeography and speciation in *Diporiphora* lizards and common geographic breaks in co-
- 605 distributed taxa. *Molecular Phylogenetics and Evolution*, 197, 108090.
- Gray, J. A., Sherratt, E., Hutchinson, M. N., & Jones, M. E. (2019a). Evolution of cranial shape in a continental-scale evolutionary radiation of Australian lizards. *Evolution*, 73(11), 2216-2229.
- 610 Gray, J. A., Hutchinson, M. N., & Jones, M. E. (2019b). Exceptional disparity in Australian agamid lizards is a possible result of arrival into vacant niche. *The Anatomical Record*, 302(9), 1536-1543.
- Harmon, L. J., Kolbe, J. J., Cheverud, J. M., & Losos, J. B. (2005). Convergence and the multidimensional niche. *Evolution*, 59(2), 409-421.
- 615 Harmon, L. J., Losos, J. B., Jonathan Davies, T., Gillespie, R. G., Gittleman, J. L., Bryan Jennings, W., Kozak, K.H., McPeck, M.A., Moreno-Roark, F., Near, T.J., Purvis, A., Ricklefs, R.E., Schluter, D., Schulte III, J.A., Seehausen, O., Sidlauskas, B.L., Torres-Carvajal, O., Weir, J.T., & Mooers, A. Ø. (2010). Early bursts of body size and shape evolution are rare in comparative data.
- 620 *Evolution*, 64(8), 2385-2396.
- Hugall, A. F., Foster, R., Hutchinson, M., & Lee, M. S. (2008). Phylogeny of Australasian agamid lizards based on nuclear and mitochondrial genes: implications for morphological evolution and biogeography. *Biological Journal of the Linnean Society*, 93(2), 343-358.
- 625 Hutchinson, G.E. (1957). Concluding Remarks. Cold Spring Harbor Symposia on Quantitative Biology, 22, 415-427.

- Kalyaanamoorthy, S., Minh, B. Q., Wong, T. K., Von Haeseler, A., & Jermin, L. S. (2017). ModelFinder: fast model selection for accurate phylogenetic estimates. *Nature methods*, 14(6), 587-589.
- 630 Keogh, J. S. (1998). Molecular phylogeny of elapid snakes and a consideration of their biogeographic history. *Biological Journal of the Linnean Society*, 63(2), 177-203.
- Liaw, A., & Wiener, M. (2002). Classification and regression by randomForest. *R news*, 2(3), 18-22.
- 635 Marshall, C. R. (2014). The evolution of morphogenetic fitness landscapes: conceptualising the interplay between the developmental and ecological drivers of morphological innovation. *Australian Journal of Zoology*, 62(1), 3-17.
- Macey, J. Robert, James A. Schulte, Allan Larson, Natalia B. Ananjeva, Yuezhao Wang, Rohan Pethiyagoda, Nasrullah Rastegar-Pouyani, and Theodore J. Papenfuss. Evaluating trans-Tethys migration: an example using acrodont lizard phylogenetics. *Systematic Biology* 49, no. 2 (2000): 233-256.
- 640 Meiri, S. (2010). Length-weight allometries in lizards. *Journal of Zoology*, 281(3), 218-226.
- Melville, J., Schulte, J. A., & Larson, A. (2001). A molecular phylogenetic study of ecological diversification in the Australian lizard genus *Ctenophorus*. *Journal of Experimental Zoology*, 291(4), 339-353.
- 645 Melville, J., Ritchie, E. G., Chapple, S. N., Glor, R. E., & Schulte II, J. A. (2018). Diversity in Australia's tropical savannas: An integrative taxonomic revision of agamid lizards from the genera *Amphibolurus* and *Lophognathus* (Lacertilia: Agamidae). *Memoirs of Museum Victoria*, 77, 41-61.
- 650 Melville, J., Date, K. S., Horner, P., & Doughty, P. (2019). Taxonomic revision of dragon lizards in the genus *Diporiphora* (Reptilia: Agamidae) from the Australian monsoonal tropics. *Memoirs of Museum Victoria*, 78, 23-55.
- Melville, J., & Wilson, S. K. (2019). Dragon Lizards of Australia: Evolution, ecology and a comprehensive field guide. Museums Victoria Publishing.
- 655 Minh, B. Q., Nguyen, M. A. T., & von Haeseler, A. (2013). Ultrafast approximation for phylogenetic bootstrap. *Molecular Biology and Evolution*, 30(5), 1188-1195.
- Minh, B. Q., Schmidt, H. A., Chernomor, O., Schrempf, D., Woodhams, M. D., Von Haeseler, A., & Lanfear, R. (2020). IQ-TREE 2: new models and efficient methods for phylogenetic inference in the genomic era. *Molecular Biology and Evolution*, 37(5), 1530-1534.
- 660 Moen, D. S. (2019). What determines the distinct morphology of species with a particular ecology? The roles of many-to-one mapping and trade-offs in the evolution of frog ecomorphology and performance. *The American Naturalist*, 194(4), E81-E95.
- 665

- 670 Pagel, M., O'Donovan, C., & Meade, A. (2022). General statistical model shows that macroevolutionary patterns and processes are consistent with Darwinian gradualism. *Nature communications*, 13(1), 1113.
- Pavanetto N, Puglielli G (2024). *funspace*: Creating and Representing Functional Trait Spaces. R package version 0.2.2, <https://CRAN.R-project.org/package=funspace>.
- Pepper, M., & Keogh, J. S. (2021). Life in the “dead heart” of Australia: The geohistory of the Australian deserts and its impact on genetic diversity of arid zone lizards. *Journal of Biogeography*, 48(4), 716-746.
- 675 Pennell, M. W., Eastman, J. M., Slater, G. J., Brown, J. W., Uyeda, J. C., FitzJohn, R. G., ... & Harmon, L. J. (2014). geiger v2. 0: an expanded suite of methods for fitting macroevolutionary models to phylogenetic trees. *Bioinformatics*, 30(15), 2216-2218.
- R Core Team. (2024). R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria. www.R-project.org
- 680 Rannala, B., Yang, Z. (2007) Inferring speciation times under an episodic molecular clock. *Systematic Biology*, 56:453-466.
- Revell, L. J. (2024). phytools 2.0: an updated R ecosystem for phylogenetic comparative methods (and other things). *PeerJ*, 12, e16505.
- 685 Schluter, D. (2000). The ecology of adaptive radiation. OUP Oxford.
- Shoo, L. P., Rose, R., Doughty, P., Austin, J. J., & Melville, J. (2008). Diversification patterns of pebble-mimic dragons are consistent with historical disruption of important habitat corridors in arid Australia. *Molecular Phylogenetics and Evolution*, 48(2), 528-542.
- 690 Simpson, G. G. (1944). Tempo and mode in evolution. Columbia University Press.
- Singhal, S., Grundler, M., Colli, G., & Rabosky, D. L. (2017). Squamate conserved loci (Sq CL): a unified set of conserved loci for phylogenomics and population genetics of squamate reptiles. *Molecular ecology resources*, 17(6), e12-e24.
- Smith, K. L., Harmon, L. J., Shoo, L. P., & Melville, J. (2011). Evidence of constrained phenotypic evolution in a cryptic species complex of agamid lizards. *Evolution*, 65(4), 976-992.
- 695 Streicher, J. W., Schulte, J. A., & Wiens, J. J. (2016). How should genes and taxa be sampled for phylogenomic analyses with missing data? An empirical study in iguanian lizards. *Systematic biology*, 65(1), 128-145.
- 700 Thompson, G., & Withers, P. (2005). Shape of western Australian dragon lizards (Agamidae). *Amphibia-Reptilia*, 26(1), 73-85.
- Tiatragul, S., Brennan, I. G., Broady, E. S., & Keogh, J. S. (2023). Australia's hidden radiation: Phylogenomics analysis reveals rapid Miocene radiation of blindsnakes. *Molecular Phylogenetics and Evolution*, 185, 107812.

- 705 Title, P. O., Singhal, S., Grundler, M. C., Costa, G. C., Pyron, R. A., Colston, T. J.,
... & Rabosky, D. L. (2024). The macroevolutionary singularity of snakes.
Science, 383(6685), 918-92.
- Wagner, P., Stanley, E. L., Daza, J. D., & Bauer, A. M. (2021). A new agamid lizard
in mid-Cretaceous amber from northern Myanmar. *Cretaceous Research*, 124,
710 104813.
- Welt, R. S., & Raxworthy, C. J. (2022). Dispersal, not vicariance, explains the
biogeographic origin of iguanas on Madagascar. *Molecular Phylogenetics and
Evolution*, 167, 107345.
- Wainwright, P. C., Alfaro, M. E., Bolnick, D. I., & Hulsey, C. D. (2005). Many-to-
one mapping of form to function: a general principle in organismal design?.
715 *Integrative and Comparative Biology*, 45(2), 256-262.
- Wright, S. (1932). The roles of mutation, inbreeding, crossbreeding and selection in
evolution. In: Proceedings of the sixth international congress of genetics, pp
356–366.
- 720 Yoder, J. B., Clancey, E., Des Roches, S., Eastman, J. M., Gentry, L., Godsoe, W.,
... & Harmon, L. J. (2010). Ecological opportunity and the origin of adaptive
radiations. *Journal of Evolutionary Biology*, 23(8), 1581-1596.
- Zhang, C., & Mirarab, S. (2022). Weighting by gene tree uncertainty improves
accuracy of quartet-based species trees. *Molecular Biology and Evolution*,
725 39(12), msac215.